

MetalSynth-Pelvis: Conditioned synthesis of osteosynthesis implants and local metal artifacts in pelvic CT scan volumes using Multi-Window Latent Diffusion

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1. Problem Statement

Automatic pelvic computed tomography (CT) analysis and volumetric segmentation depend on sufficiently annotated data. Metal artifacts can obscure anatomical boundaries near implants, motivating evaluation of segmentation robustness in this region. CTPelvic1K (Liu et al., 2021) includes a CLINIC-metal subset whose limited bone annotation constrains supervised evaluation. Its name does not establish that every case contains pelvic osteosynthesis, nor that other subsets are free of metal or foreign objects. The local audit will therefore determine implant types, eligible training anatomy and annotated evaluation cases through three-dimensional and multiplanar review. The magnitude of peri-implant segmentation degradation must be measured in this study rather than assumed from the dataset description.

While generative models based on diffusion processes excel at synthesizing visually plausible medical anomalies, they structurally fail when applied to metallic implants. State-of-the-art lesion synthesis models, such as DiffTumor (Chen et al., 2024), DiffBoost (Zhang et al., 2025), CLAIM (Ramzan et al., 2026), and LGESynthNet (Jacob et al., 2026), are designed around bounded inpainting for deformable biological tissues. They implicitly assume non-rigidity and strictly restrict intensity alterations to the inside of the object's mask. This is physically incompatible with metal implants, which are rigidly geometric and produce severe attenuation profiles (beam hardening, photon starvation, and high-amplitude streaking) that propagate globally far beyond the implant's boundaries, destroying the signal of the surrounding soft tissue.

Physics-based metal artifact simulation provides a reference for appearance generation. This thesis adopts the hybrid simulation protocol of Peters et al. (2025), implemented with CatSim within XCIST (De Man et al., 2007; Wu et al., 2022), as its physical comparison protocol. Their phantom validation supports the evaluated simulation settings; it does not establish perfect physical coherence or validate pelvic osteosynthesis in three dimensions. Their training data use randomized metal placement, whereas their scoring benchmark uses expert placement. The unresolved placement problem is therefore scalable, anatomically constrained placement, rather than scalable metal insertion itself.

Even if automated placement were achieved, mere anatomical containment is insufficient; a valid simulation must reflect true surgical admissibility. Traditional deterministic surgical planning pipelines (Liu et al., 2025) attempt to optimize a single "perfect" trajectory. Yet real-world clinical data reveal a different reality. Grading iliosacral screws on the four-level cortical-breach scale of Smith et al. (2006), Zwingmann et al. (2009) report two distinct postoperative-CT distributions: grade 0 in 69% of navigated placements (grades 1–3: 15%, 8%, 8%) against grade 0 in 40% of conventional placements (37%, 11.5%, 11.5%). These are two surgical-technique populations that differ significantly, not one aggregate range; this work therefore targets the ordinal distributions themselves rather than a single imperfection rate. Sacral level also warrants separate treatment on anatomical grounds. Using multiplanar CT measurements, Gardner et al. (2010) report mean minimum iliosacral cross-sectional areas at S1 and S2 of 346 and 109.3 mm² in normal sacra, compared with 222 and 220.1 mm² in dysmorphic sacra. The approximately equal areas in the latter group support measuring each level on the individual anatomy rather than assuming a universal corridor-size hierarchy. These measurements concern an uninjured side and ideal central trajectories; they do not describe clinical malposition frequencies. In a retrospective clinical series, van den Bosch et al. (2002) reported postoperative neurologic complaints in 6 of 31 patients with the lower screw in S2, versus 1 of 49 with both screws in S1 ($p = 0.01$). These are patient-level outcomes by screw configuration, not screw-malposition rates by level; the authors also identify a learning-curve bias associated with the change in placement practice. This clinical association supports retaining S1/S2 as a sampler variable, but supplies neither a causal level effect nor an ordinal cortical-breach prior for S2. The ordinal benchmark from Zwingmann et al. (2009) applies to S1; S2 placements are evaluated descriptively for geometry and breach grades without claiming clinical distribution matching. The premise that no canonical pose exists is not only inferred from surgical error rates; it is supported anatomically. Measuring the orientation of the bony surfaces that bound the corridor across seventeen cadaveric pelves, Ziran et al. (2007) report inter-specimen coefficients of variation of 7–25% for most surfaces, rising to 43%, 47–52% and, for the orientation of the superior S1 ala in the sacral frontal plane, 97–140%. A corridor that varies this much between individuals cannot be served by one fixed trajectory. Training a diagnostic network solely on "perfect" deterministic trajectories will inevitably cause it to fail when encountering the true variance of clinical errors. Therefore, a generative sampler must not optimize for perfection, but instead sample from the measured ordinal distribution of clinical malpositions, bounded by pelvic density representations (Arand et al., 2019) and by osseous-corridor viability. For the latter, McLaren et al. (2021) supply the operational element that a qualitative safe-zone description (Ramadanov and Zabler, 2025) does not: a reproducible corridor-diameter procedure over 433 pelves, yielding transiliosacral angular tolerances of $1.53 \pm 0.57^\circ$ at S1 and $1.02 \pm 0.33^\circ$ at S2. Their ≥ 10 mm corridor criterion is not an empirically validated safety threshold and is not treated as one here: Kaiser et al. (2014), from whom it is taken, describe it as a chosen conservative value and justify it dimensionally, as clearance of 1–2 mm around a 6.3–8 mm screw. It is therefore used as a geometric clearance convention rather than as an inherited clinical standard. Kaiser et al. (2014) additionally supply what a corridor diameter alone does not: a reference frame computable on a CT volume, obtained by reformatting along the sacral axis perpendicular to the superior end plate of S1, with corridor angulation measured coronally against the line joining the iliac crests and axially against the line joining the posterior iliac spines, and a usable-length rule requiring at least 5 mm of clearance to the cortex on either side. The pose distribution sampled within that frame remains an explicit construction of this work.

Recognized Gap. Current attempts to bridge the metal-artifact domain shift are bottlenecked by three mutually exclusive paradigms: physical simulators do not supply scalable anatomically constrained placement, biological diffusion models cannot generate rigid global streaks, and deterministic surgical planners ignore the statistical reality of clinical malposition. To the best of the author's knowledge, there is no diffusion-based method that overcomes these barriers by combining: **(C1)** Non-biological rigid implant geometries, which avoid the threshold-dependent size distortion of fixed-HU metal segmentation (over-coverage in simulated slices, Xie et al., 2024; fragmentation of pelvic screws in the local audit); **(C2)** A surgically constrained placement sampler that accurately emulates the population distribution of clinical malpositions; and **(C3)** The transfer of multi-window HU encoding from artifact *removal*, where it originates, to artifact *synthesis*, coupled with a generation band (B_δ) extended beyond the implant mask so that global streaking and beam hardening are modelled outside the metal itself. The multi-window framework is not claimed as novel here: it is established in metal-artifact-reduction work, and Wang et al. (2025) attribute it in turn to earlier multiple-window learning. What is claimed is its use for generation, together with B_δ , for which no published precedent quantifying a peri-implant band was found. This is the gap this work aims to fill.

2. Research Question

What improvement does a surgical admissibility-constrained (bone density and cortical containment) 3D placement sampler, a multi-window Latent Diffusion Model (LDM) conditioned on the rigid implant geometries and an extended generation band (B_δ), have in terms of distributional physical-coherence of the synthesized pelvic osteosynthesis implants and their local artifacts, when compared with both naive copy-paste insertion and the adopted physics-based simulation protocol of Peters et al. (2025)? Downstream segmentation impact is deliberately excluded from this study and is stated as out of scope below.

3. Hypothesis

The synthetic generation of medical anomalies and metallic implants using a rigid geometry and extended spatial boundaries, HU-conditioned diffusion model and whose placement is surgically enforced via a density-based placement sampler is more physically-coherent than unconstrained generative baseline methods and on the same footing as the adopted explicit physics-based simulation protocol (Peters et al., 2025). Placement coherence is quantified by Surgical Admissibility of Placement (SAP), the single metric introduced by this work, which emulates the measured ordinal distribution of real-world implant placement. Appearance coherence is quantified with the metrics of the adopted protocol under their published names, not with newly coined ones. Ablations reveal the non-redundant role of the multi-window encoding and of the extended boundary generation.

4. Objectives

General Objective. Design, implement and evaluate a physics-and-anatomy-aware latent diffusion pipeline with rigid surgical constraints and extended HU-based conditioning, capable of generating physically coherent metallic implants intended, in future work, to improve the robustness of downstream segmentation networks.

1. **Validation of a multi-window representation (Compulsory Go/No-Go):** Ensure HU reconstruction error (HU to multi-window to VAE to HU again) within the bone tissue below a 25 HU threshold (Mean Absolute Error (MAE)). This will serve as a guarantee that the intensity gradients that drive downstream applications are preserved.
2. **Design of surgically constrained placement sampler:** propose position-orientation 3D pose of a parametric rigid screw model constrained by bone density maps (Arand et al., 2019), cortical boundaries with a 5 mm clearance rule, and osseous-corridor viability (McLaren et al., 2021), expressed in the CT-computable reference frame of Kaiser et al. (2014) (sacral-axis reformat; coronal and axial corridor angulation against iliac-crest and posterior-iliac-spine landmarks). The sampler must yield a distribution of placements emulating the two measured ordinal malposition distributions of Zwingmann et al. (2009), conditioned on surgical technique within S1, rather than a single artificial "optimized" trajectory. Sacral level remains a geometric conditioning variable, anatomically motivated by the segment-specific corridor measurements of Gardner et al. (2010); S2 results are reported descriptively, because the clinical comparison of van den Bosch et al. (2002) concerns neurologic complaints by screw configuration and does not provide an ordinal S2 placement distribution. Sacral-morphology phenotypes are deliberately not used as a stratifying variable: their reported effect on safe-zone size is not consistent across studies, and the sampler instead measures corridor geometry directly on each volume.
3. **Design of a conditioned synthesizer:** 2.5D ControlNet-guided Latent Diffusion Model (Stable Diffusion 1.5 backbone). A crucial detail in the formulation is that the synthesis mask must be extended with a spatial band B_δ ($\sim 12\text{mm}$) over the implant boundary, to allow a more explicit manifestation of the streaking artifacts and beam hardening that occur outside of the implant mask in the generated image.
4. **Formalization of the placement metric and adoption of an inherited appearance protocol:** Formulation of Surgical Admissibility of Placement (SAP) as the only metric introduced by this work, evaluated with Wasserstein-1 distances against the measured ordinal distributions. Appearance is evaluated with the metrics of the adopted protocol under their published names — bone integrity, metal integrity and streak amplitude (Peters et al., 2025) — and the previously proposed BFC and ISC are withdrawn, so that inherited measures are not renamed as contributions.

Explicitly out of scope. Downstream segmentation evaluation (Dice, HD95) is excluded from this study. Two auditable conditions drive the exclusion rather than time alone: within the locally available cohort only a minority of the metal-bearing volumes carry verified bone annotation, and only part of the published collection is held locally, so a downstream result would rest on a cohort too small and too partially annotated to support the claim. The synthesis arms and the placement sampler are therefore evaluated on physical coherence and surgical admissibility alone; downstream utility is left as future work.

5. Expected Results

Goal	Metric(s)	Expected Outcome
Surgical Admissibility (Only novel metric)	SAP: four-level Cortical Breach Grade (Smith et al., 2006); density-zone fraction; corridor viability (McLaren et al., 2021) in the reference frame of Kaiser et al. (2014).	Distribution matching: Wasserstein-1 distance shows generated placements approaching the two measured ordinal distributions of Zwingmann et al. (2009), conditioned on technique within S1. S2 geometry and breach grades are reported descriptively, without a clinical ordinal target.
Appearance Coherence (Inherited protocol, published names)	Bone integrity; metal integrity; streak amplitude (Peters et al., 2025). Inversion of each metric for synthesis is documented, and the 0–4 scale is not carried over.	Lower profile discrepancy than naive copy-paste insertion, with streak amplitudes statistically comparable to, or better than, the adopted physics-based protocol.
Representation Viability (Standard Encoding Eval)	MAE in HU (Round-trip)	Error in the bone tissues strictly below 25 HU threshold. Log-compressed metal ranges preserve the structural topology.
Preservation outside B_δ (No-damage check)	RMSE, SSIM against the original CT, restricted to the complement of B_δ .	No measurable change where none is intended; deviation inside B_δ is the intended signal and is not scored as error.
Ablation validation (Constraint Verification)	All metrics above	Incremental ordering shows their non-redundant contribution. Degradation when B_δ band and multi-window HU encoding are removed.

6. Datasets

Primary Dataset. CTPelvic1K (Liu et al., 2021) will be used as the source of patient anatomy and real artifact observations. The locally available dataset6 and dataset7 volumes are inventoried separately from the full published collection; the CLINIC-metal label alone does not establish implant type or annotation availability. *Strict Isolation Rule.* No patient may contribute to both training and evaluation. The implant geometries consumed by the proposed sampler and by the physical simulation arm are parametric and are not derived from any evaluation volume. Duplicate-volume screening and patient grouping are required to enforce this separation.

Implant geometry source. Implants are modelled as parametric rigid screws rather than extracted from the metal-bearing volumes. Shaft diameter is drawn from the calibres reported for iliosacral fixation, 6.5–8.0 mm (Gardner et al., 2010) and 6.3–8 mm (Kaiser et al., 2014). Screw length is not taken from a fixed published range: it is bounded on each volume by the measured corridor under the usable-length rule of Kaiser et al. (2014). Features without a textual source in the reviewed literature (thread, cannulation, head and washer) are simplified to a smooth cylinder, and this simplification is declared as a limitation. The real implants of the metal-bearing subset do not enter the geometry bank; they serve as a descriptive contrast for placements, as an appearance reference for synthesis, and as a test–retest control where the same implant was imaged twice. The local audit motivates this choice: screws extracted from these volumes with a fixed HU threshold appear fragmented, with shaft diameters below the published calibres, and the extracted geometry changes between acquisitions of the same implant.

Physical comparison protocol. The hybrid simulation and evaluation protocol of Peters et al. (2025), also associated with the AAPM CT-MAR Challenge (Haneda et al., 2025), is adopted instead of an independently validated reimplement of XCIST/CatSim. Challenge data remain excluded from model training, and the scoring benchmark is reserved for final evaluation. Reproduction will document the code revision, configuration, reconstruction settings and access conditions. Both synthesis arms will use the same source anatomy and implant poses; a no-metal simulation/reconstruction control will quantify changes introduced by the physical arm alone. The published protocol is two-dimensional and evaluates MAR: extending it to pelvic osteosynthesis synthesis requires an explicit adaptation and does not inherit its validation automatically. RMSE and SSIM against the original CT will assess preservation outside the generation band, where changes are not intended; artifact realism within the affected region requires separate evaluation. No unverified numerical acceptance thresholds or MAR score calibration are adopted here.

Field limitation: corridor geometry has only been characterized in intact pelves. This is not a limitation of the present cohort but of the available literature. Every source that publishes sacral corridor geometry measured it in pelves without implants and without fracture: Kaiser et al. (2014) excluded scans with implants obscuring the lumbosacral junction, McLaren et al. (2021) used uninjured scans, and Ziran et al. (2007) worked on cadaveric specimens and warn that fracture, displacement and soft-tissue injury would alter what is visible. The reference frame adopted here rests on landmarks at the lumbosacral junction, which is precisely where streaking concentrates, so whether those landmarks remain locatable in metal-bearing volumes is not answered by any published source. It was therefore quantified on the local cohort. Landmarks were located by a heuristic on each volume, and a landmark region was counted as contaminated when its dark-streak or metal fraction exceeded the maximum observed in 69 metal-free patients of the same cohort. Among 65 patients with orthopaedic metal, all five landmarks were found within the field of view in 57. After a clinician reviewer (one surgeon, otorhinolaryngology), blinded to the automatic audit, confirmed the S1 level on every patient, the frame was computable in 48 patients and free of contamination at all five landmark regions in 29. Losses were driven more by field-of-view truncation of the iliac crests (7 patients) and by landmark localisation than by artifact contamination; in the 17 patients with a screening-detected iliosacral screw, the S1 level was correct in all and its region was contaminated in 3. The frame is therefore computed on metal-bearing volumes where it is computable; the handling of field-of-view-truncated cases is assessed separately, and the heuristic localisation of the iliac crests and posterior iliac spines has not been clinically reviewed.

Local cohort audit. Training eligibility requires manual review of three-dimensional views and multiplanar CT slices confirming absence of metal and foreign objects. HU screening only prioritizes this review, including cases from dataset6. Confirmed metal cases are candidates for testing, with pelvic osteosynthesis distinguished from other implants and external objects. Exact volume duplicates and repeated patients must be resolved before splitting. Final cohort sizes remain pending this audit.

Why downstream evaluation is excluded. The audit itself supplies the reason. Only a minority of the metal-bearing volumes carry verified bone annotation, and only a fraction of the published collection is available locally, so a Dice or HD95 result would rest on a cohort too small and too partially annotated to support a robustness claim. The metal-bearing subset used here is nonetheless the de facto clinical evaluation set for pelvic metal-artifact work, having been used as such in recent published work (Wang et al., 2025). Where a clean reference image does not exist for these volumes, appearance realism is assessed against real artifact observations rather than against an unavailable ground truth.

7. References

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